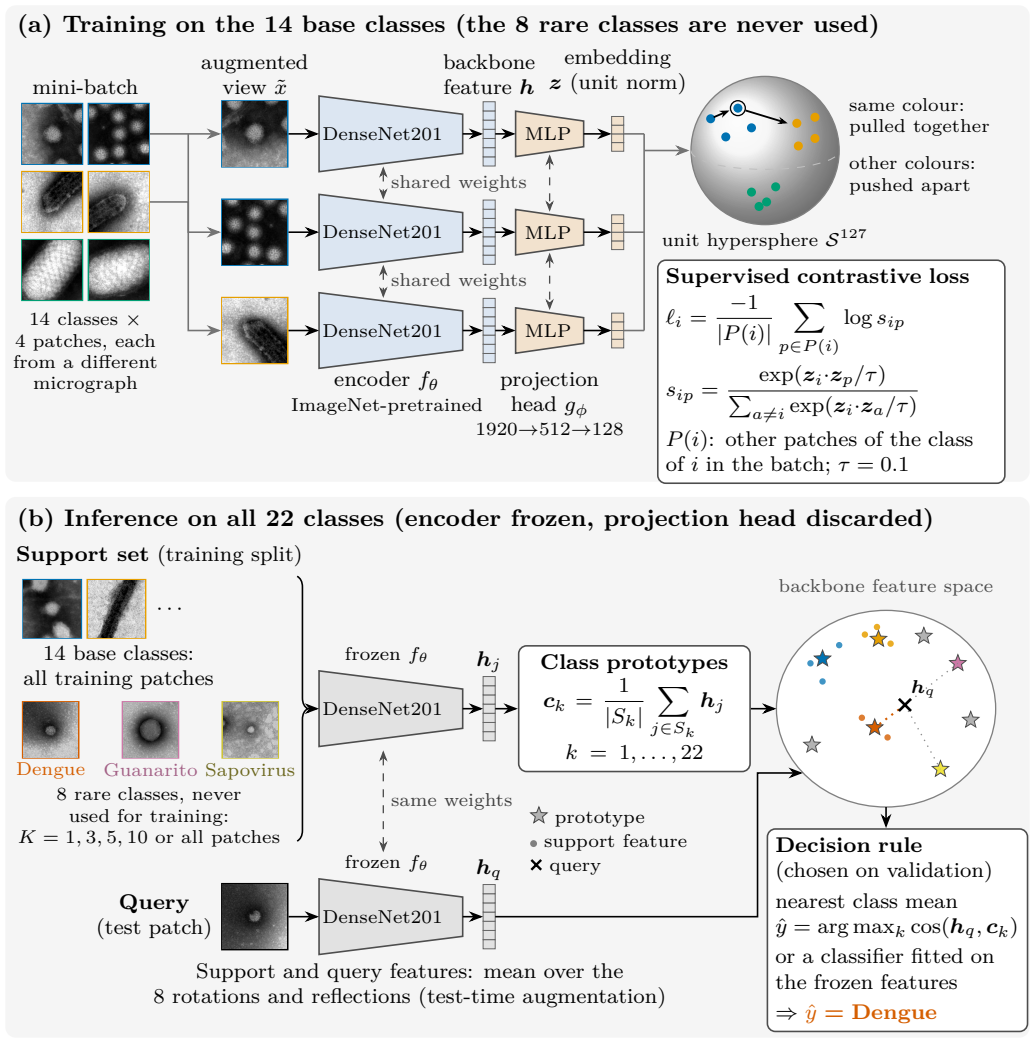

(a) Training on the 14 base classes (the 8 rare classes are never used)
mini-batch
augmented view $\tilde{x}$
backbone feature $\boldsymbol{h}$
embedding $\boldsymbol{z}$ (unit norm)
DenseNet201
MLP
shared weights
shared weights
same colour: pulled together
other colours: pushed apart
unit hypersphere $\mathcal{S}^{127}$
14 classes × 4 patches, each from a different micrograph
encoder f_θ
ImageNet-pretrained
projection head g_ϕ
1920→512→128
Supervised contrastive loss
$\ell_i = \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log s_{ip}$
$s_{ip} = \frac{\exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_p / \tau)}{\sum_{a \neq i} \exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_a / \tau)}$
$P(i)$: other patches of the class of i in the batch; $\tau = 0.1$
(b) Inference on all 22 classes (encoder frozen, projection head discarded)
Support set (training split)
14 base classes: all training patches
Dengue
Guanarito
Sapovirus
8 rare classes, never used for training: $K = 1, 3, 5, 10$ or all patches
frozen f_θ
same weights
frozen f_θ
Class prototypes
$\boldsymbol{c}_k = \frac{1}{|S_k|} \sum_{j \in S_k} \boldsymbol{h}_j$
$k = 1, \ldots, 22$
backbone feature space
prototype
support feature
query
Query (test patch)
Decision rule
(chosen on validation)
nearest class mean
$\hat{y} = \arg\max_k \cos(\boldsymbol{h}_q, \boldsymbol{c}_k)$
or a classifier fitted on the frozen features
$\Rightarrow \hat{y}$ = Dengue
Support and query features: mean over the 8 rotations and reflections (test-time augmentation)